higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE

INDUSTRIAL ELECTRONICS N4

1 APRIL 2019

This marking guideline consists of 6 pages.

Copyright reserved

QUESTION 1

1.1 1.1.1 Maximum power transfer
1.1.2 Superposition theorem
1.1.3 Kirchhoff's second law (Voltage law)
1.1.4 Thévenin's theorem

(4 × 1) (4)

1.2 $I = 30/(18 + 11 + 7)$
$= 0,8333$ A

$V_{th} = IR$
$= 0,8333 \times 18$
$= 14,994$ V

$R_{th} = (18 \times 18)/(18 + 18)$
$= 9\ \Omega$

$I = V_{th}/(R_{th} + R_{12\Omega})$
$= 14,994/(9 + 12)$
$= 0,714$ A (8)

[12]

QUESTION 2

2.1 The ratio of the reactance of the inductor or capacitor to its equivalent resistance (2)

2.2 $Q = X_L/R$
$= 2\ 500/1\ 000$
$= 2,5$ (1)

2.3 $Z_1 = 23 + j18$
$= 29,206 \angle 38,047$

$Z_2 = 40 - j31$
$= 50,606 \angle -37,776$

$Z_3 = 55 + j30$
$= 62,65 \angle 28,61$

$Z_S = 23 + j18 + 40 - j31$
$= 63 - j13$
$= 64,327 \angle -11,66$

$Z_T = (64,327 \angle -11,66 \times 62,65 \angle 28,61)/(23 - j13 + 55 + j30)$
$= 4\ 030,087 \angle -16,95/79,831 \angle 12,295$
$= 50,483 \angle 4,655$

$I_T = V_T/Z_T$
$= 240 \angle 0/50,483 \angle 4,655$
$= 4,754 \angle -4,655$ A (8)

[11]

Copyright reserved

QUESTION 3

3.1 When each electron combines with a hole in the P-type material each acceptor has an extra electron. (2)

3.2 3.2.1 The positive side of the external voltage repels the holes in the P-type material and causes them to drift across the junction.

3.2.2 Its positive side attracts the electrons from the N-type material and its negative side attracts the holes from the P-type material.

(2 × 2) (4)

3.3 $I = Is(e^{qv/kT} - 1)$ $qv/KT = 24,615$

$= 45 \times 10^{-6}(e^{24,615} - 1)$

$= 2\ 205\ 658,247$ A (3)

3.4 $R = KT/qi$

$= (1,38 \times 10^{-23} \times 305)/(0,805 \times 1,6 \times 10^{-19})$

$= 0,0327$

$= 32,679$ mA (3)

[12]

QUESTION 4

4.1 4.1.1 The ratio of the number of turns in the primary to the number in the secondary

4.1.2 A measure of the smoothness of the DC output of a rectifier and a ratio of the rms value of the AC ripple to the average value of the component

(2 × 2) (4)

4.2 4.2.1 $V_{dc} = 280((4 \times 60 \times 120 \times 10^{-6} \times 250)/(4 \times 60 \times 120 \times 10^{-6} \times 250+1))$

$= 245,854$ V

4.2.2 $\lambda = 1/(4 \times \sqrt{3} \times 60 \times 120 \times 10^{-6} \times 250)$

$= 0,0802$

(2 × 2) (4)

4.3 $V_{dc} = 178/(178 + 150 \times 200)$

$= 108,537$ V (2)

4.4
- Increased ripple with load current
- Lower output due to voltage drop across the resistor
- Regulation worse than the LC-input filter (Any 2 × 1) (2)

[12]

QUESTION 5

5.1
- Cut-off region: the collector and emitter junction are both reverse-biased resulting in almost no collector current.
- Saturation region: both junctions are forward-biased.
- Active region: the base-emitter junction is forward-biased while the base-collector junction is reverse-biased. (3 × 2) (6)

5.2
- Produces a much larger output signal than a single class-A amplifier
- Improves the efficiency (2)

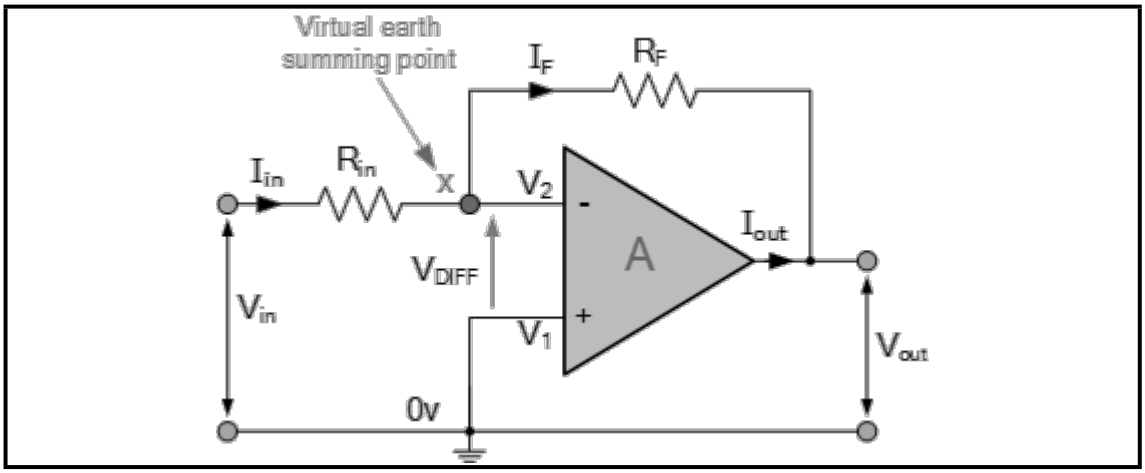
5.3 The difference lies in the construction of the gate terminal – instead of the reverse PN-junction used with the JFET, the MOSFET has a capacitor-like connection at its gate consisting of a metal gate contact, silicon dioxide dielectric and a conducting channel. (3)

[11]

QUESTION 6

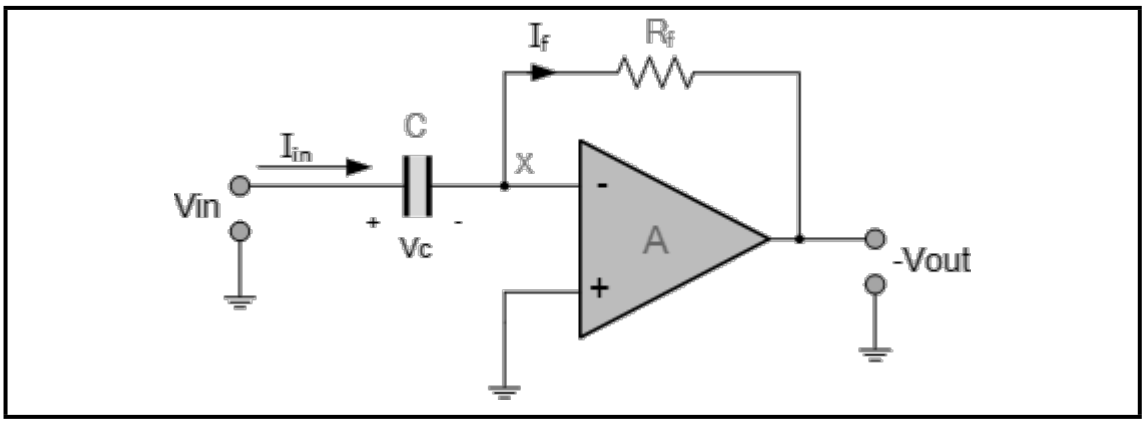
6.1
- Noninverting input
- Inverting input (2)

6.2



(4)

6.3



(4)

[10]

QUESTION 7

7.1 There is no current through the thyristor. (2)

7.2
- Relay control
- Time-delay circuits
- Regulated power supplies
- Static switches
- Motor controls
- Heater controls (Any 3 × 1) (3)

7.3 7.3.1 Compares the output with the reference

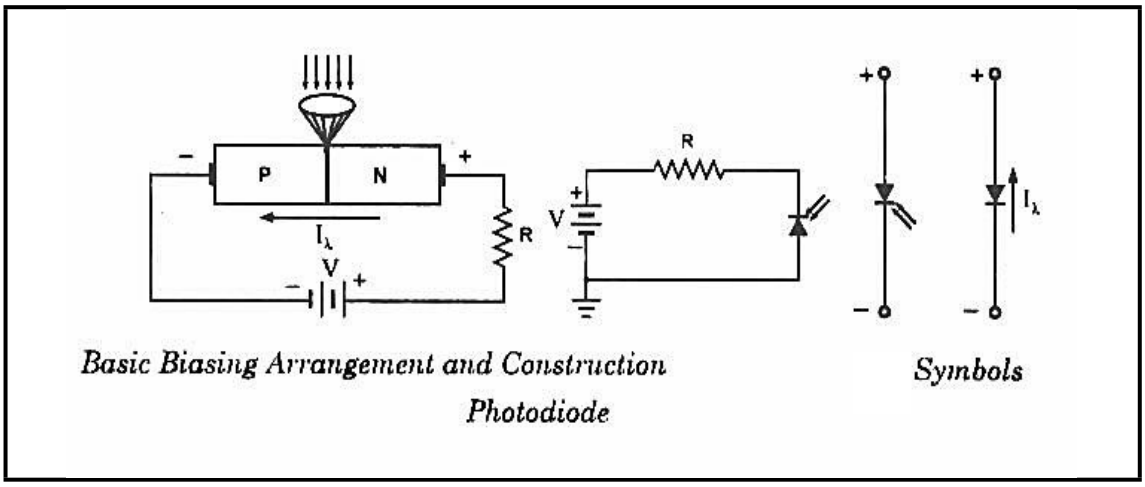
7.3.2 Network consisting of a detecting element

7.3.3 Adjusts the physical quantity on which the value of the controlled condition depends in response to a signal from comparator (3 × 2) (6)

[11]

QUESTION 8

8.1



(5)

8.2 To convert AC output to DC output that is sometimes used for recorders, indicators and controllers (2)

8.3 To allow the phototransistor to be operated as a normal bipolar transistor with or without being exposed to light (3)

8.4 It will increase the base current and create a greater increase in the collector current. (2)

[12]

QUESTION 9

9.1
- AC and DC volts
- Frequency
- Phase difference between signals
- Analyse waveforms (Any 3 × 1) (3)

9.2 9.2.1 To measure any signal displayed on the screen one must first select a reference line.

9.2.2 It determines the brightness of the spot on the display.

9.2.3 Used to measure a steady trace, to delay the time-base sweep until the input signal is at the beginning of a new cycle

(3 × 1) (3)

9.3
- Sine waveforms
- Square waveforms
- Triangular waveforms (3)

[9]

TOTAL: 100

Copyright reserved